

Revision Date 19-Apr-2018

WAI1 - AGHS - OSHA

Revision Number 1

**1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING****Product Identifier****Product Name** TRO Reagent 2**Product No** TROREC2RD**Pure substance/mixture** Mixture**Relevant identified uses of the substance or mixture and uses advised against****Recommended Use** Use as laboratory reagent**Uses advised against** No Information available**Manufacturer, Importer, Supplier** Thermo Fisher Scientific©  
Water and Lab Products  
22 Alpha Road  
Chelmsford, MA 01824, USA  
1-978-232-6000**E-mail address** [info.water@thermo.com](mailto:info.water@thermo.com)**Made in** USA**Emergency Telephone** 24 Hour Emergency Phone Number  
CHEMTREC®  
Within USA and Canada: 1-800-424-9300  
Outside USA and Canada: 1-703-527-3887  
(collect calls accepted)

## 2. HAZARDS IDENTIFICATION

### Classification

#### OSHA Regulatory Status

This chemical is not considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

### Label Elements

#### Emergency Overview

**Appearance** Clear

**Physical State** Liquid

**Odor** None

### Precautionary Statements

#### Hazards not otherwise classified (HNOC)

No information available

#### Other Information

No information available

## 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component        | CAS-No    | Weight % |
|------------------|-----------|----------|
| Water            | 7732-18-5 | >90.0%   |
| Potassium Iodide | 7681-11-0 | 1 - 10%  |
| Sodium Hydroxide | 1310-73-2 | <0.1%    |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

## 4. FIRST AID MEASURES

### First aid measures

#### General Advice

Use first aid treatment according to the nature of the injury. Get medical attention immediately if symptoms occur. Show this safety data sheet to the doctor in attendance.

#### Eye Contact

Rinse thoroughly with plenty of water, also under the eyelids. Get medical attention.

#### Skin Contact

Wash off immediately with soap and plenty of water for at least 15 minutes. Take off contaminated clothing and shoes immediately. In case of skin reactions, consult a physician.

#### Inhalation

Move to fresh air. If breathing is difficult, give oxygen. Get medical attention if symptoms occur.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Do not induce vomiting. Call a physician or Poison Control Center immediately.

#### Self-Protection of the First Aider

Use personal protective equipment. See section 8 for more information. Do not use

mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

**Most important symptoms and effects, both acute and delayed**

**Most important symptoms and effects** No information available

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically

**5. FIRE-FIGHTING MEASURES**

**Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable Extinguishing Media**

No information available

**Specific Hazards Arising from the Chemical**

No information available.

**Explosion Data**

**Sensitivity to Mechanical Impact** None

**Sensitivity to Static Discharge** None

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**6. ACCIDENTAL RELEASE MEASURES**

**Personal Precautions, Protective Equipment and Emergency Procedures**

**Personal Precautions** Use personal protective equipment. For further specification, refer to section 8 of the SDS.  
Evacuate personnel to safe areas.

**Environmental Precautions** Collect spillage. See Section 12 for additional ecological information.

**Methods and Material for Containment and Cleaning Up**

**Methods for Containment** Prevent further leakage or spillage if safe to do so.

**Methods for Cleaning Up** Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

**7. HANDLING AND STORAGE**

**Precautions for Safe Handling**

**Handling** To avoid risks to human health and the environment, comply with the instructions for use  
Wear personal protective equipment  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Ensure adequate ventilation, especially in confined areas

**Conditions for Safe Storage, Including any Incompatibilities**

**Storage** Keep container tightly closed in a dry and well-ventilated place  
Store at room temperature in the original container  
Keep away from direct sunlight

Incompatible Products No information available

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control parameters

#### Exposure Guidelines

| Component                     | ACGIH TLV                    | OSHA PEL                                                           | NIOSH IDLH                                                 |
|-------------------------------|------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|
| Potassium Iodide<br>7681-11-0 | TWA: 0.01 ppm                | -                                                                  | -                                                          |
| Sodium Hydroxide<br>1310-73-2 | Ceiling: 2 mg/m <sup>3</sup> | (Vacated) Ceiling: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>Ceiling: 2 mg/m <sup>3</sup> |

#### Appropriate engineering controls

Engineering Measures Showers  
Eyewash stations  
Ventilation systems

#### Individual protection measures, such as personal protective equipment

**Eye/face Protection** Wear chemical splash goggles and face shield. If splashes are likely to occur, wear:  
Face-shield.

**Skin and Body Protection** Wear protective gloves/clothing.

**Respiratory Protection** None under normal use conditions. In case of inadequate ventilation wear respiratory protection.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

Physical State Liquid  
Appearance Clear  
Odor None  
Odor Threshold No information available  
pH 11  
PH Range No information available

| Property                     | Values                   | Remarks • Method |
|------------------------------|--------------------------|------------------|
| Melting point/freezing point | No information available |                  |
| Boiling Point/Range          | 100 °C / 212 °F          |                  |
| Flash Point (High in °C)     | N/A                      |                  |
| Evaporation Rate             | No information available |                  |
| Flammability (solid, gas)    | No information available |                  |
| Flammability Limit in Air    |                          |                  |
| Upper flammability limit:    | No information available |                  |
| Lower flammability limit:    | No information available |                  |
| Vapor pressure               | No information available |                  |
| Vapor Density                | No information available |                  |
| Specific Gravity             | 1.016g/mL                |                  |
| Water Solubility             | Soluble in water         |                  |
| Solubility in other solvents | No information available |                  |
| Partition coefficient        | No information available |                  |
| Autoignition Temperature     | -                        |                  |
| Decomposition Temperature    | No information available |                  |
| Kinematic viscosity          | No information available |                  |

|                      |                          |
|----------------------|--------------------------|
| Dynamic viscosity    | No information available |
| Explosive Properties | No information available |
| Oxidizing Properties | No information available |

**Other Information**

|                  |                          |
|------------------|--------------------------|
| Softening Point  | No information available |
| Molecular Weight | No information available |
| VOC Content(%)   | No information available |
| Density          | No Information available |
| Bulk Density     | No information available |

## 10. STABILITY AND REACTIVITY

**Reactivity**

No Information available

**Chemical Stability**

Stable under normal conditions

**Possibility of Hazardous Reactions**

None under normal processing

**Conditions to Avoid**

Extremes of temperature and direct sunlight.

**Incompatible Materials**

No information available

**Hazardous Decomposition Products**

Thermal decomposition can lead to release of irritating gases and vapors.

## 11. TOXICOLOGICAL INFORMATION

**Information on likely routes of exposure**

|              |                          |
|--------------|--------------------------|
| Inhalation   | No information available |
| Eye Contact  | No information available |
| Skin Contact | No information available |
| Ingestion    | No information available |

**Information on Toxicological Effects**

|          |                          |
|----------|--------------------------|
| Symptoms | No information available |
|----------|--------------------------|

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                           |
|------------------------|---------------------------|
| Sensitization          | No information available  |
| Mutagenic Effects      | No information available  |
| Carcinogenicity        | No information available. |
| Reproductive Effects   | No information available  |
| STOT - single exposure | No information available  |

STOT - repeated exposure No information available

Aspiration hazard No information available

**Numerical measures of toxicity - Product Information**

The following values are calculated based on chapter 3.1 of the GHS document .

**12. ECOLOGICAL INFORMATION**

**Ecotoxicity**

1.66% of the mixture consists of components(s) of unknown hazards to the aquatic environment

| Component                     | Freshwater Algae | Freshwater Fish                                        | Water Flea |
|-------------------------------|------------------|--------------------------------------------------------|------------|
| Sodium Hydroxide<br>1310-73-2 | -                | LC50: = 45.4 mg/L, 96h static<br>(Oncorhynchus mykiss) | -          |

**Persistence and Degradability**

No information available

**Bioaccumulation/ Accumulation**

No information available

**Mobility**

No information available.

**Other adverse effects**

No information available

**13. DISPOSAL CONSIDERATIONS**

**Waste treatment methods**

**Waste Disposal Methods** Disposal should be in accordance with applicable regional, national and local laws and regulations.

**Contaminated Packaging** Improper disposal or reuse of this container may be dangerous and illegal.

| Component                     | CAWAST             |
|-------------------------------|--------------------|
| Sodium Hydroxide<br>1310-73-2 | Toxic<br>Corrosive |

**14. TRANSPORT INFORMATION**

**DOT** Not regulated

**ICAO** Not regulated

**IATA** Not regulated

**IMDG/IMO** Not regulated

**15. REGULATORY INFORMATION**

**International Inventories**

|               |          |
|---------------|----------|
| USINV         | Complies |
| CANINV        | Complies |
| EINECS/ELINCS | Complies |
| ENCS          | Complies |
| IECSC         | Complies |
| KECL          | Complies |
| PICCS         | Complies |
| AICS          | Complies |

**Legend:**

USINV/ TSCA - United States Toxic Substances Control Act Section 8(b) Inventory  
CANINV/ DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List  
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
ENCS - Japanese Existing and New Chemical Substances  
IECSC - Chinese Inventory of Existing Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances  
AICS - Australian Inventory of Chemical Substances

**U.S. Federal Regulations**

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications. Under the amended regulations at 40 CFR 370, EPCRA 311/312 Tier II reporting for the 2017 calendar year will need to be consistent with updated hazard classifications.

**CWA (Clean Water Act)**

| Component                     | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|-------------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Sodium Hydroxide<br>1310-73-2 | 1000 lb                     | -                      | -                         | X                          |

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component                     | Hazardous Substances RQs | CERCLA EHS RQs | RQ                                        |
|-------------------------------|--------------------------|----------------|-------------------------------------------|
| Sodium Hydroxide<br>1310-73-2 | 1000 lb                  | -              | RQ 1000 lb final RQ<br>RQ 454 kg final RQ |

**U.S. State Regulations**

**California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

**U.S. EPA Label Information**

No information available

**16. OTHER INFORMATION**

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|                            |                                |
|----------------------------|--------------------------------|
| <b>Prepared By</b>         | Regulatory Affairs             |
| <b>Prepared For</b>        | Thermo Fisher Scientific Inc.© |
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| <b>Revision Date</b>       | 19-Apr-2018                    |
| <b>Reason for revision</b> | Initial Release.               |

**Disclaimer**

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**End of Safety Data Sheet**